UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Bike Frame Topology Optimization

Mechanical Science & Engineering

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ME 498 Fall 2025 – Prof. Golecki

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Introduction





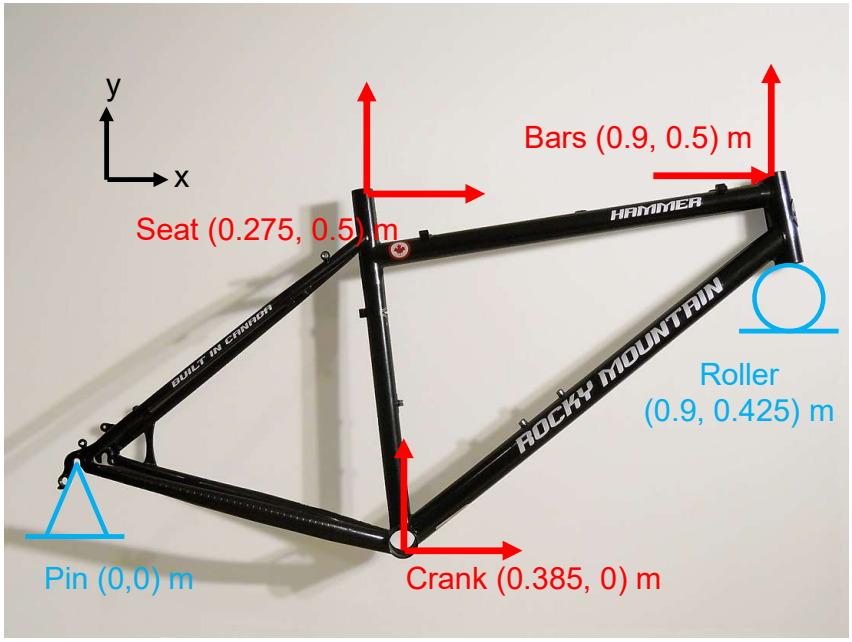
Introduction





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Assumptions, FBD



- 2D Frame
- Only direct loads (loads directly onto bike frame vs onto components)
- No out of plane forces
- No chain tension force
- Pin and Roller Constraints

Configurations

A. Cruising

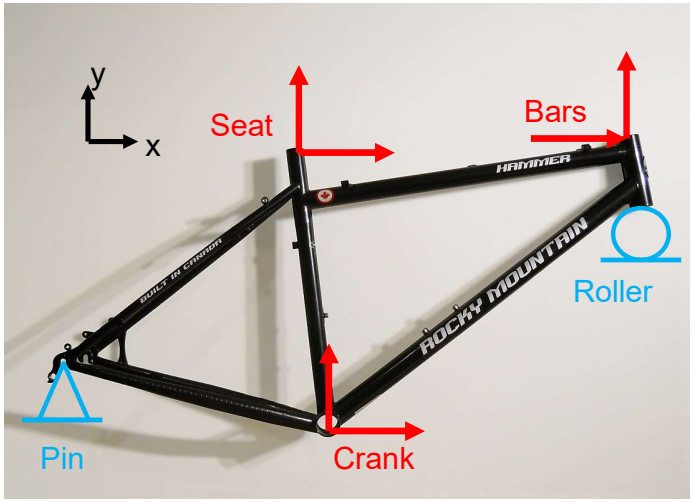


B. Climbing



C. Braking





Force values for cruising, climbing, and braking load configurations			
Value	A. Cruising	B. Climbing	C. Braking
$F_{crank,x}$ **[N]**	150	200	500
$F_{crank,y}$ **[N]**	-500	-1500	-500
$F_{seat,x}$ **[N]**	0	0	0
$F_{seat,y}$ **[N]**	-500	-25	-200
$F_{bar,x}$ **[N]**	-100	-100	-1000
$F_{bar,y}$ **[N]**	-100	-100	-200

Optimization Statement

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & C(\mathbf{x}) = \boldsymbol{f}^{\mathrm{T}}\mathbf{u}(H\mathbf{x}) \\ \text{s.t.} \quad & K(H\mathbf{x})\mathbf{u} - \boldsymbol{f} = \mathbf{0}, \\ & \textstyle\sum_{e=1}^{n_e}(\rho t A_e \cdot H x_e) - 0.3 m_{\max} \leq 0, \\ & 10^{-3} \leq x_e \leq 1.0, \end{aligned}$$

where:

$\mathbf{x}$ is a vector of n element densities,

H is the hat-filter matrix,

$C(\mathbf{x})$ is the compliance value,

$\boldsymbol{f}$ is the force vector,

$\mathbf{u}(H\mathbf{x})$ is the displacement vector,

$K(H\mathbf{x})$ is the SIMP stiffness matrix,

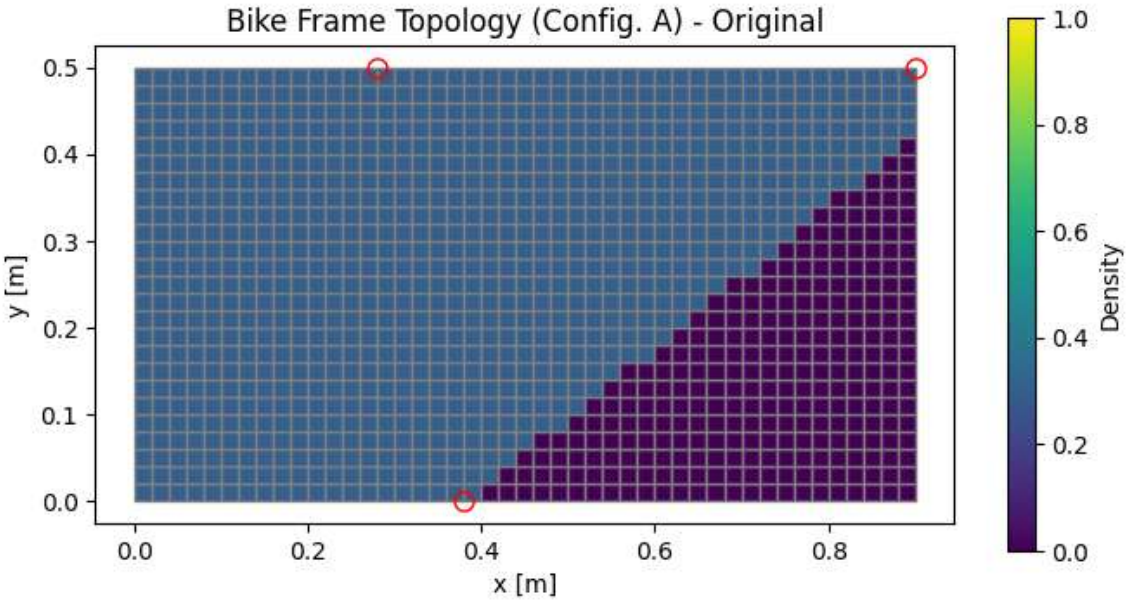
ρ is the material density, t is the plate thickness, A_e is the element area,

and $m_{\max}$ is the maximum area if the entire design domain were filled.

Problem Classification:

Nonlinear, convex, continuous (digitally discrete, functionally continuous) topology optimization

Setup

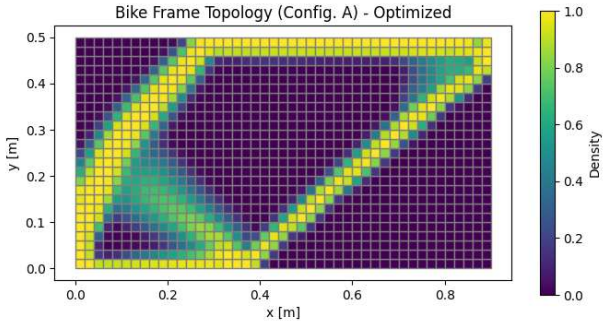


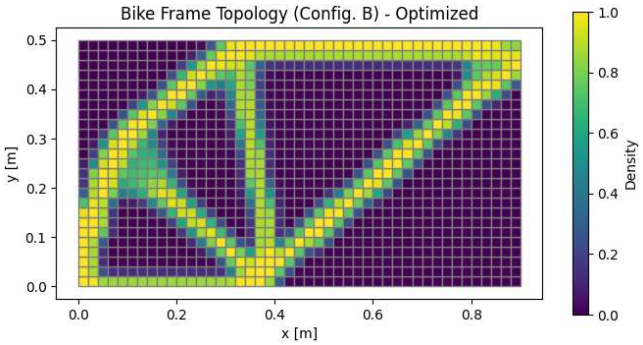
Mesh grid size: (45, 25)
Aluminum Cost and Material Properties

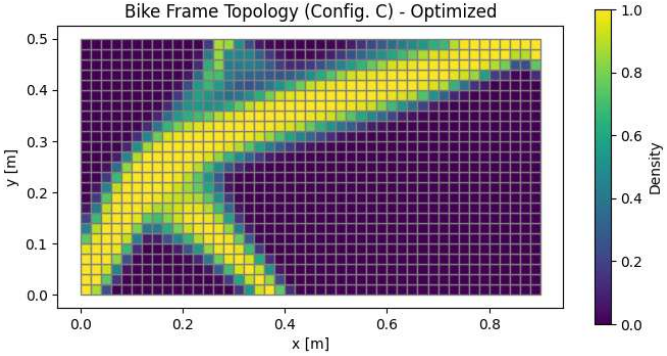
Density Filter Radii	
Configuration	r_{min} [m]
Cruising	0.023
Climbing	0.027
Braking	0.025

Results

Cost: $48.26 (18.56 kg)





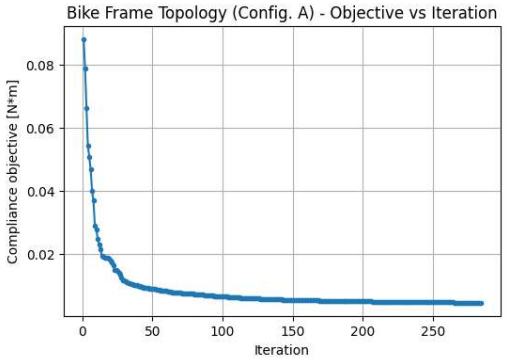


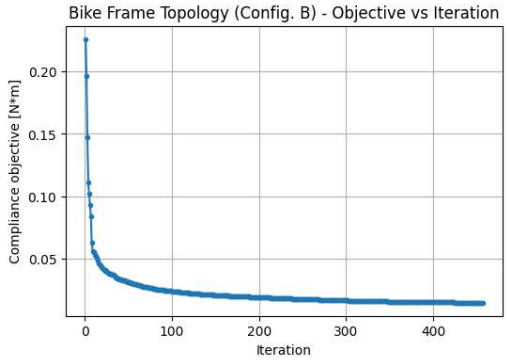


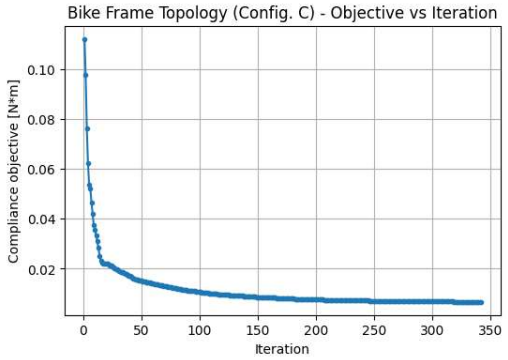




Results







Final Compliances	
Configuration	C [$\times 10^{-3} N \cdot m$]
Cruising	4.62
Climbing	14.95
Braking	6.70

Conclusions

Key Takeaways

- Generated optimal bike-frame topologies for three distinct load configurations.
- Topology optimization is only a tool

Future Work

- Explore alternative load configurations
- Introduce more realistic passive void regions (e.g., chain clearance) and enforce required material zones.
- Incorporate more complex and accurate loading conditions to reduce simplifying assumptions in the model.